

LECTURE 5: INFECTIONS OF THE EYES and EAR

Eye infections

Conjunctivitis

It's an inflammation of the conjunctiva mostly caused by bacteria or viruses. Conjunctivitis can also be caused by chemicals, allergy, fungi, or parasites.

Thus, the etiologic differentiation is crucial for correct diagnosis and treatment to avoid possible loss of vision.

- Viral Conjunctivitis

Adenovirus is the common

virus

Pharyngoconjunctival fever (swimming pool conjunctivitis occur in outbreaks in summer camps by types 3, 4, 7).

Conjunctivitis and keratoconjunctivitis by types 8 and 19 in epidemics

Severe ocular irritation, copious watery discharge, and pre auricular lymph node (LN) swelling.

Treatment: Usually self-limited but contagious. A topical corticosteroid is used if it's severe.

Adenovirus: dsDNA, icosahedral symmetrical, non-enveloped, Fibers protruding from the capsid

infection, the adenovirus enters human epithelial cells, replicates, and causes host cell death. Transition to latent infection involves lymphoid tissue, where the virus may remain dormant.

- Bacterial Conjunctivitis

-Staph, Strep, Hib, Pseudomonas:

Features: Foreign body sensation and purulent discharge. Diagnosis: Gram stain + culture (if severe).

Treatment: Empiric antibiotic drops and ointment.

-N. gonococcus: Features: Copious purulent discharge. It may lead to corneal perforation and blindness.

Dx: Gram stain shows Gram - intracellular diplococci.

Treatment: Emergent IM ceftriaxone; oral ciprofloxacin or ofloxacin.

-Chlamydial Features: C. trachomatis A-C is the common cause of chronic conjunctivitis worldwide.

Presenting with recurrent epithelial keratitis in childhood, with corneal scarring, and entropion.

Diagnosis: Giemsa stain and culture for Chlamydia.

Treatment: Azithromycin, tetracycline, or erythromycin for 3-4 weeks.

Chlamydia: Obligate intracellular pathogens with three sp.

A. trachomatis serovars A, B, and C are associated with ocular infections,

serovars D through K are associated with conjunctivitis, urethritis, cervicitis, and pneumonia;

The infectious agent is the elementary body form. The elementary bodies bind to receptors on host epithelial cells to initiate infection

Metabolically active, replicating intracellular forms are referred to as reticulate bodies. Keratitis

Features: Corneal infection caused by foreign body, contact lens and trauma, etc. Hazy cornea with central ulcerations and adjacent stromal abscesses.

Tx: Refer to an ophthalmologist immediately.

Fungal keratitis Features: Usually seen in farmers and immunodeficient patient. Fungal growth on the cornea with multiple stromal abscesses.

Retinitis

-CMV retinitis Features: common viral infection of the eyes with AIDS and immunodeficiency. Painless retinitis without conjunctivitis or keratitis. Fundoscopy: Fluffy/granular retinal lesion with opacity or hemorrhage.

Corneal scraping staining (Tzanck test) usually shows multinuclear giant cells. Tx: Refer to an ophthalmologist immediately.

HSV retinitis Features: Painful, with conjunctivitis or keratitis.

Fundoscopy: Retinal pallor, ulcerations or necrosis.

Corneal scraping staining usually shows multinuclear giant cells. Tx: Refer to an ophthalmologist immediately.

Orbital cellulitis

Features: Peri- or post orbital septum infection; more common in children.

Sudden fever, proptosis, painful eyes, decreased eye movement, and red swollen eyelids.

Tx: Systemic antibiotics.

Endophthalmitis

severe inflammation of the inside of eye from a bacterial or fungal infection.

Candida the most common cause

after eye surgeries or eye is penetrated by an object.

severe eye pain, vision loss, blurry vision

pus or discharge, sensitivity to bright

lights Treatment depends on cause

Chalazion

a granulomatous inflammation of the meibomian gland due to

obstruction changing from painless or painful swelling to a firm nodule

on the lid.

It often co-exists with blepharitis.

Tx: 1. Cleaning and compression initially incision and drainage if

necessary. Hordeolum (Stye)

an abscess caused by Staph-aureus infection. Painful, tender swelling localized to the eyelid.

Ear infections

Otitis Media

It's an infection of the middle ear between the eustachian tube and the tympanic membrane. The most common microorganisms are Pneumococcus, followed by Hib and Moraxella catarrhalis. This is the same organism type as in bronchitis and sinusitis; viruses cause most other cases.

Essentials of diagnosis

1. History of ear pain, fever, and decreased hearing.
2. Physical Examination (P/E) reveals a red, bulging tympanic membrane with immobility and loss of the light reflex, with air insufflations.

Membrane perforation is a rare complication.

Treatment

Antibiotics: Oral amoxicillin. amoxicillin-clavulanate is used if there is resistance. 2nd- or third-generation cephalosporins.

Azithromycin or clarithromycin for allergic patients.

If there are multiple recurrences or if there is no response to multiple antibiotics, tympanocentesis should be performed.

Complications

Membrane perforation and hearing loss:

Recurrent otitis media: Otitis media with effusion

Mastoiditis: Redness and tenderness over the mastoid bone

Cholesteatoma: A pocket of sq epithelium in the tympanic membrane.

Meningitis or brain abscess:

Labyrinthitis: Vertigo, nystagmus, tinnitus, hearing loss, and vomiting. Otitis externa

Otitis Externa It's also known as "Swimmer's ear", inflammation of the skin lining the ear canal and surrounding soft tissue.

Continuous wetting of the ear canal and local trauma increase the risk. Pseudomonas-A and Staph-aureus are the most common pathogens.

Essentials of diagnosis: 1. Ear pruritus and pain exacerbated by moving the ear canal, which is inflamed, swollen, and macerated with purulent discharge.

2. Systemic symptoms may or may not be present.

Topical antibiotics: Eardrops with polymyxin B, neomycin, and hydrocortisone are very effective. Dicloxacillin is good for acute infection.

Patient with diabetes (DM) is at increased risk for malignant otitis externa and skull osteomyelitis, requiring IV antibiotics in hospital.

Prevention: For frequent swimming activities, use ear spray with dilute alcohol immediately after swimming